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Rediscovering the Latent Dimensions of Personality with Large Language Models as Trait Descriptors

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Abstract

Assessing personality traits using large language models (LLMs) has emerged as an interesting and challenging area of research. While previous methods employ explicit questionnaires, often derived from the Big Five model of personality, we hypothesize that LLMs implicitly encode notions of personality when modeling next-token responses. To demonstrate this, we introduce a novel approach that uncovers latent personality dimensions in LLMs by applying singular value decomposition (SVD) to the log-probabilities of trait-descriptive adjectives. Our experiments show that LLMs "rediscover" core personality traits such as extraversion, agreeableness, conscientiousness, neuroticism, and openness without relying on direct questionnaire inputs, with the top-5 factors corresponding to Big Five traits explaining 74.3% of the variance in the latent space. Moreover, we can use the derived principal components to assess personality along the Big Five dimensions, and achieve improvements in average personality prediction accuracy of up to 5% over fine-tuned models, and up to 21% over direct LLM-based scoring techniques.

1 Introduction

Over the past decades, researchers in personality psychology have investigated the viability of a general, descriptive model that yields systematic categorization of individual differences in personalities. These efforts culminated in one of the most widely-accepted taxonomies—the *Big Five* model [18, 11] with extraversion, agreeableness, conscientiousness, neuroticism, and openness as five principal dimensions of personality traits—that has been consistently cross-validated in various studies with different human populations and assessment methods [24, 17, 6, 12]. The core underlying principle behind the Big Five is the *lexical hypothesis*, which asserts that the fundamental characteristics of personality are likely encoded in natural language, often as single words [3, 9, 30, 18]. The five factors were derived from analyses of terms that people frequently use to describe personalities of themselves and others, resulting in lists of trait descriptive adjectives (TDAs) [27, 10, 11] that form the basis of psycholexical analyses and have influenced questionnaire-based assessments employed to-date, such as the Big Five Inventory (BFI) [33] and NEO-FFI [24, 6].

More recently, advancements in large language models (LLMs) have sparked investigations of whether these models are capable of recognizing and expressing personality traits as humans [16, 4, 32, 29], with potential applications in automatic text-based personality recognition [23, 19, 36, 5, 35]. As LLMs are trained on increasing amounts of text corpora, the expectation is that these models could correlate linguistic features reflected in samples of text to the personality dimensions of the authors [28, 26]. Prior work explores direct prompting, *i.e.* asking to directly predict the Big Five trait scales, or prompting to complete questionnaires [16, 32]. While these work report that personality prediction may benefit from the use of LLMs, the dependence on direct prompts and questionnaires has potential risks of unreliable and prompt-sensitive results [34, 13, 7].

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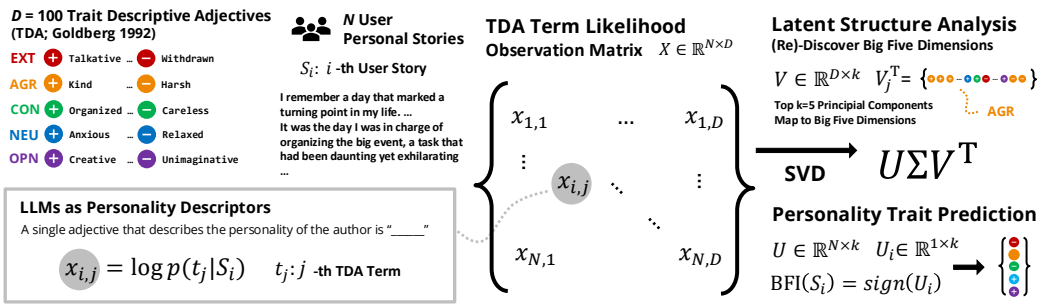

Figure 1: Overview of our approach. Given a set of personal stories and a list of TDAs, a language model computes the likelihood of each trait term describing the author of the story, through which we construct an observation matrix of such log-probabilities. Singular Value Decomposition (SVD) is then applied to this matrix, which yields (1) the loading matrix V that captures the latent dimension structure and (2) the factor matrix U that explains each story author's personality spectrum in the projected low-dimensional space. Results are compared against findings from psychometric literature and binary Big Five labels of authors, respectively.

In this work, we first revisit the lexical hypothesis and the analyses that led to the discovery of the Big Five model: does LLMs' description of personality encode latent structures that align with the Big Five? We examine whether the same principle dimensions reported in human studies emerge when LLMs are prompted to describe personalities without imposing pre-defined taxonomies, thereby allowing natural, unrestricted depiction of personality traits. Specifically, we condition language models so as to measure the likelihoods of each term in the Goldberg's 100-TDA [10] in describing the author, over a set of synthetic personal stories. These likelihoods, computed as log-probabilities, are analyzed using singular value decomposition (SVD) to uncover underlying personality factors.

The uncovered factors are then used to predict the polarity of Big Five traits of the authors of unseen stories. Overall, our results demonstrate not only that LLMs encode a latent structure similar to the Big Five model, with the five principal factors explaining 74.3% of total variance, but also our approach to personality trait predictions outperform previous methods, by up to 5% over models fine-tuned on Big Five trait classification and by up to 21% over direct LLM-scoring approaches.

2 Using Language Models as Personality Descriptors

In this work, we hypothesize that LLMs encode personality traits implicitly within their latent spaces. More explicitly, we argue that we can analyze and extract personality traits present in personal stories by carefully querying the LLM and analyzing the log-probabilities of trait-descriptive adjectives (TDAs) within the next token prediction. Here we ask: is it possible to recover traditional categories of personality from the principal components of the resulting token likelihood matrix?

TDA Assessment Prompt

Following is a personal story.
Essay: ***(Personal Story)***

Question: Based on this essay, describe the personality of the author.
Answer with a single adjective.

Answer: A single adjective that describes the personality of the author is "

Figure 2: A question-answer format prompt that asks for a single adjective describing the personality of the author of the given personal story. A quotation mark is adopted at the prompt suffix to guide the model to complete the sentence with an adjective. For each adjective in the TDA, we measure the log-probability of the model completing the prompt with the given adjective.

We present the primary steps of our approach in Figure 1. First, we measure the likelihood of a set of personality trait terms describing the story reflecting each individual author, *i.e.* calculate the log-probabilities of the 100 adjectives in the TDA curated in [10]. In doing so, we prompt LLMs with a personal story as illustrated in Figure 2. Then, we build an observation matrix where each row contains the log-probabilities of TDA for each user. These log-probabilities can capture the usage frequency of adjectives varies in real life, based on the input prompt. For instance, adjectives associated with extraversion are more commonly used than those related to conscientiousness, leading to differences in latent factor levels [9]. Therefore, this subtle difference is captured in log-probabilites, allowing the method to recover the five principal axes corresponding

to each trait in the Big Five model. Further details on the measurement process can be found in Appendix A.

After building the observation matrix, we perform singular value decomposition (SVD) on the matrix to identify the top-5 factors. Based on the SVD results, we analyze both the loading matrix and the factor matrix (Section 3). We then report the classification results derived from the factor matrix and compare these with the findings from the previous psychometric literature [17, 9, 10, 11].

2.1 Re-Discovering the Big Five Structure

We start with measuring the log-probabilities of the trait terms in describing each of the users and construct the observation matrix $X \in \mathbb{R}^{N \times D}$, where N is the number of users and D is the number of trait adjectives in Figure 1. We observe that the log-probabilities are biased by the frequency of usage of trait adjectives, as shown in Figure 3, and accordingly apply zero-centering. SVD is applied to the matrix X, resulting in $\tilde{X} = U\Sigma V^\top$—in this decomposition, $U \in \mathbb{R}^{N \times k}$ is the **factor matrix** that represents the latent features of personality traits of each story author, $\Sigma \in \mathbb{R}^{k \times k}$ contains the singular values, and $V \in \mathbb{R}^{D \times k}$ is the **loading matrix** that defines how each trait adjective relates to the latent factors. We compare the factor matrix with factor loading identified in [10] to examine whether the principal components uncovered by the factor matrix resembles the Big Five dimensions.

2.2 Personality Trait Prediction

Each element of the factor matrix U_{ij} represents the extent of the *i*-th story's association with the *j*-th latent factor. Specifically, the sign of the element predicts the binary personality trait (e.g., 'extraverted' or 'introverted') of the story author, and for evaluation, we compare these polarities against the actual personality trait labels. Since our approach is based on an *unsupervised* method, the full procedure does not require explicit labels of personality traits. When labels are available, we can further enhance our analysis by incorporating supervised learning. Specifically, we choose Lasso regression using the log-probability features to predict the personality traits since Lasso regression performs feature selection by identifying the most informative trait adjectives.

3 Experiments

We benchmark our methods against baselines on `PersonaLLM` dataset [16]. `PersonaLLM` dataset is generated by `gpt-4-0613` [1], where `gpt-4-0613` is prompted with Big Five traits and asked to generate stories consistent with the prescribed personality. More details about the dataset are in Appendix B. We consider a suite of LLMs including Meta Llama 3 and Llama 3.1 model families [2] and Mixtral 8x22B [15, 25]. We present the analysis results that uncover the Big Five structure in Section 3.1 and compare trait prediction accuracy with baselines in Section 3.2.

3.1 The Big Five Structure

The sum of the top-5 squared singular values from SVD accounts for 74.3% of the total variance in the log-probability measurements, indicating that it sufficiently captures the variance. The detailed statistics of singular values are provided in Appendix C.1. We compare elements of principal components with factor loadings from the literature [10]. Specifically, identifying which adjectives have large elements in each principal component establishes the correspondence between principal components and personality factors. For example, since adjectives that have high loading in the extraversion factor (e.g., 'energetic', 'bashful', 'reserved') also have large elements in the first principal component, the first principal component relates to the extraversion trait. We identify the one-to-one correspondence between principal components and Big Five traits; the top-5 principal components correspond with extraversion, openness, agreeableness, neuroticism, and conscientiousness traits, respectively. Detailed steps for finding the correspondence are given in Appendix C.2.

3.2 Personality Trait Prediction

U matrix of SVD can be interpreted as the personality trait scales of each story author. More specifically, the element of U, U_{ij}, represents the scale of the j-th Big Five trait of the i-th user. By taking the signs of the predicted scales and comparing them with binary personality labels, we measure